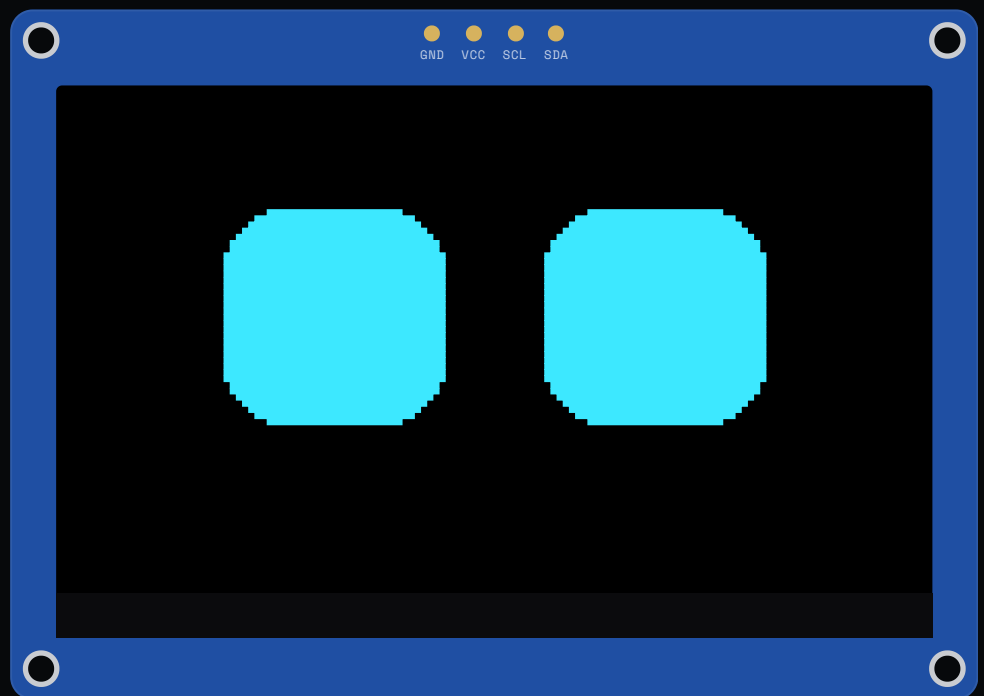


DESKTOP-PET

USER GUIDE AND TECHNICAL NOTES

A small face that lives on the desk. Two rounded eyes on a 0.96 inch OLED glance around, blink, cheer up when you pet it and fall asleep when you ignore it. Connected to the PC it becomes the face of Amino, my voice assistant. It keeps the time, shows the weather and runs timers, dances to your music, codes and games along with you, counts down your selfies, and keeps a secret behind its right cheek.



BOARD

WeAct Black Pill, STM32F401CC or F411CE

DISPLAY

SSD1306 OLED, 128 × 64 pixels, I2C

INPUT

KEY, a touch pad, an RFID card

LINKS

USB, and WiFi through an ESP-01

WHAT'S INSIDE

GUIDE

- 01 Meet the pet**
What it is and how it's connected

- 02 One button**
Tap, double tap, hold

- 03 Left to itself**
Moods, habits and the faces it can pull

- 04 Amino's face**
Listening, thinking, speaking

- 05 Music and films**
Headphones, dancing, bars and popcorn

- 06 Coding and gaming**
Glasses and paws, a controller and sparks

- 07 Selfies**
A countdown, and a flash on the real shutter

- 08 The card**
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- 09 Screens and cards**
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- 10 Things to try**
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INSIDE

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- 12 Wiring**
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From the speakers to 16 bars and a beat

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Its own time and weather, and the link to the PC

- 18 Building and uploading**
PlatformIO, the HID bootloader, flash layout

- 19 When something's off**
Troubleshooting and a map of the code

ABOUT THE PICTURES

Every screen in this handbook is a real capture. The pet was asked for its frame buffer 28 times a second over USB (the `SN` command, chapter 15) and each 1,024-byte frame was drawn back pixel for pixel. Where a figure shows a strip, the frames are in order and the times are measured from the first one. The time, the weather and the timers are example values sent to the pet for the pictures.

MEET THE PET

Desktop-pet is a face for the desk. It runs on a WeAct Black Pill, a thumb-sized board with an STM32 chip on it, and draws on a 128 × 64 OLED about the size of a postage stamp. You touch it through one button, called KEY, or a touch pad under the top of its head.

On its own it behaves like a small animal: it looks around, blinks, fidgets, enjoys being petted and dozes off when nobody pays attention. Connected to the PC it also follows what you're doing: it dances to your music, eats popcorn through films, puts on glasses while you code and picks up a controller when you play. It is the face of Aminal, my voice assistant, and a small dashboard for the time, the weather, timers and reminders.

On the PC a small bridge program does the talking. It starts when you log in and runs whether Aminal is open or not, so the pet follows the PC all day, and Aminal adds its listening, thinking and speaking on top.

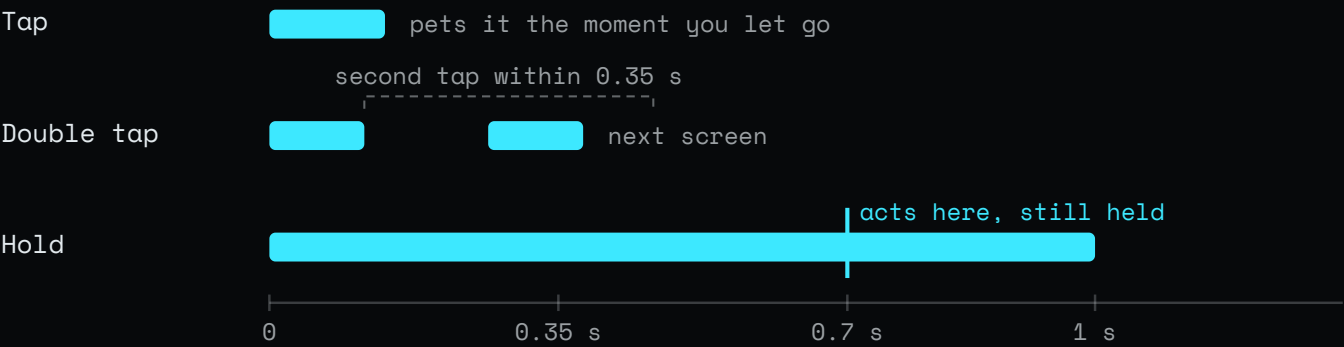
HOW IT'S CONNECTED	WHAT IT DOES
USB cable to the PC	Everything. The fastest link, always used when it's plugged in.
WiFi, on a charger	Everything, over the air. The bridge finds it about 20 to 30 seconds after it powers up.
A charger, PC off	Still a pet. Its WiFi module fetches its own time and weather.

ONE BUTTON

KEY is the small button beside the USB socket. The other one, NRST, restarts the board. What a press does depends on what is on the screen.

PRESS	ON THE FACE	ON ANY OTHER SCREEN
Tap	Pets it: happy for 2.5 seconds	Next screen
Three taps	Falls in love: two beating hearts. Each tap within 1.5 s of the last.	Three screens on
Double tap	Pets it, then flips to the next screen. Second tap within 0.35 s.	Two screens on
Hold, 0.7 s	Back to the face	On the timer: pause or carry on. Anywhere else: back to the face.

A tap also puts away a card. The screens come round in this order: face, clock, weather, timer (only while one is running) and status. A screen you flip to goes back to the face by itself after 20 seconds.



LEFT TO ITSELF

The pet is never quite still. Every 1.5 to 4 seconds it glances somewhere new, and it blinks every 2.5 to 6 seconds. Every 15 to 40 seconds it also does one small habit, picked at random: a good look left and then right, a little hop, a suspicious squint, or a double blink.



0.00 s



2.03 s



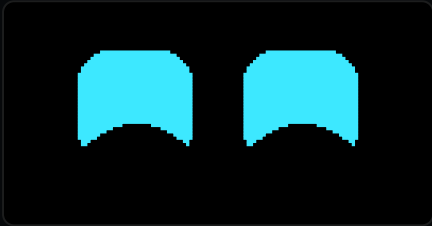
3.22 s



14.00 s

Idle

Glancing right, blinking, looking up to the left, then down. Four moments from 15 seconds of doing nothing.



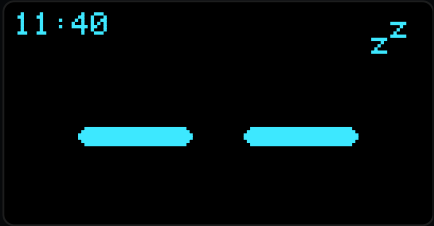
Happy

Tap once. The eyes lift and curve into arcs for 2.5 seconds.



In love

Tap three times, each within 1.5 s of the last. Two hearts beat for 2.5 seconds.

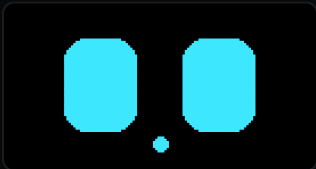


Asleep

Ignore it for 30 seconds, or 10 after 11 pm. The eyes droop to slits, z's float up and the time shows in the corner.

FACES AMINAL CAN ASK FOR

Aminal can make the pet pull any of seven faces for as long as it likes, 3 seconds unless it says otherwise: happy, love, surprised, sad, angry, wink and sleepy. The pet also looks surprised by itself when a timer runs out or a film suddenly gets loud.



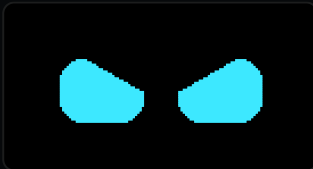
Surprised

Tall eyes and a small round mouth.



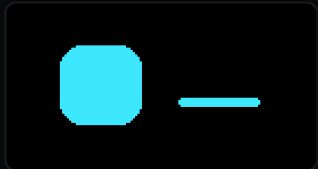
Sad

The outer corners droop.



Angry

The inner corners pull down.

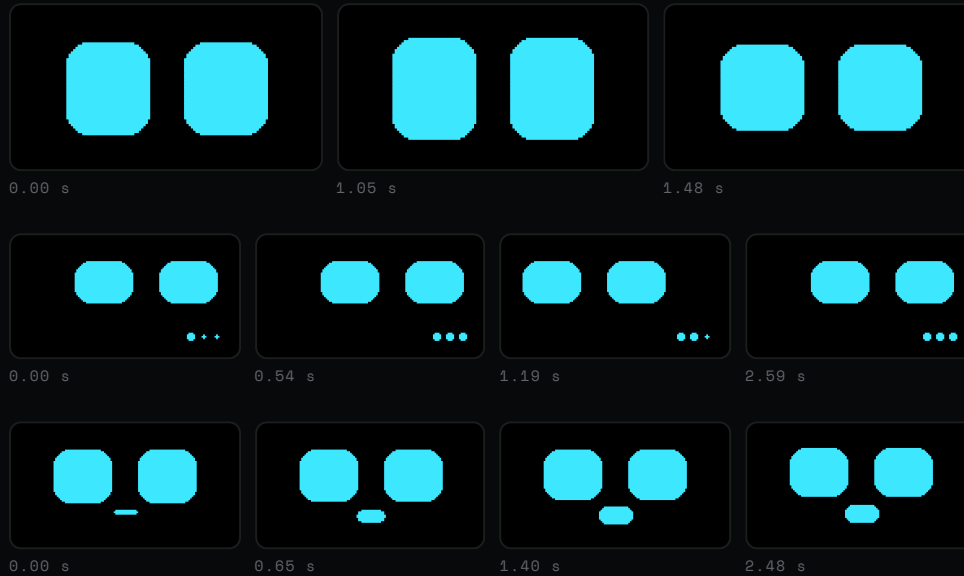


Wink

One eye closes to a line.

AMINAL'S FACE

While Aminoal runs, the pet shows what it is doing. The bridge (chapter 11) sends Aminoal's state and the loudness of the voice up to 20 times a second, so the eyes keep up with speech.



Listening

Eyes wide open, swelling with the loudness of your voice.

Thinking

Looking up and off to one side, switching sides every second or so, while three dots fill in.

Speaking

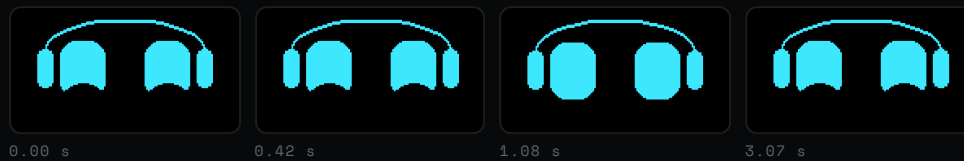
The eyes lift to make room for a mouth that opens with Aminoal's voice.

IT FOLLOWS THE CONVERSATION

- Start talking to Aminoal while the clock or the weather is up, and the pet flips back to the face to listen.
- Ask for the weather or a timer, and it shows that screen while Aminoal answers.
- Answers that carry a reminder, mail or what's next on the calendar arrive as a card (chapter 09).
- When Aminoal connects, the pet greets it with a happy face. When Aminoal closes, the pet waits 8 seconds for another message and then carries on by itself.

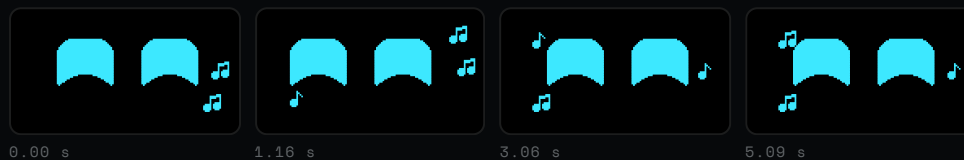
MUSIC AND FILMS

When music plays, the pet joins in with one of three moods, picked at random and changed every 25 to 45 seconds. It rarely picks the same mood twice in a row.



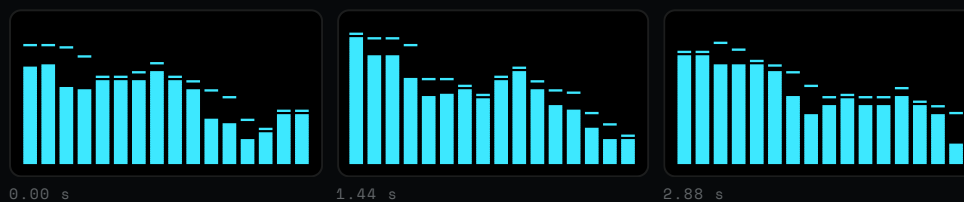
Headphones

Headphones on, eyes shut and happy, nodding on every beat. Every nine seconds or so it opens its eyes for a look around (third frame).



Dancing

Sways from side to side and hops on the beat. A note floats up on every second beat.

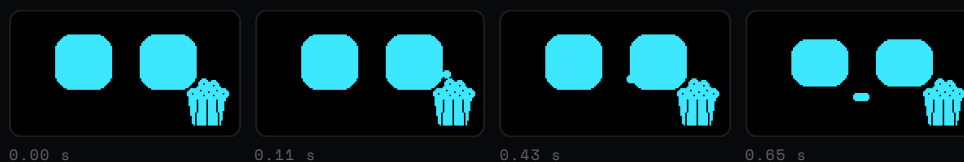


Visualizer

The whole screen turns into 16 bars of what the speakers are playing, low notes on the left, with peak marks that fall back slowly. Captured with test bars sent over USB.

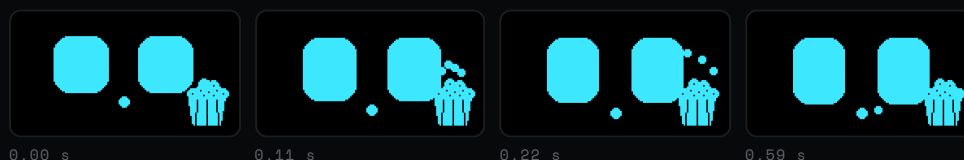
FILMS

Put on Netflix, Prime Video or Hotstar and it looks up at the screen with a striped bucket of popcorn.



Popcorn

Every 3.5 to 7 seconds it flicks a kernel out of the bucket, catches it and chews.



A loud scene

When the sound suddenly jumps, so does the pet: surprised eyes, and four kernels fly out of the bucket.

HOW IT TELLS THEM APART

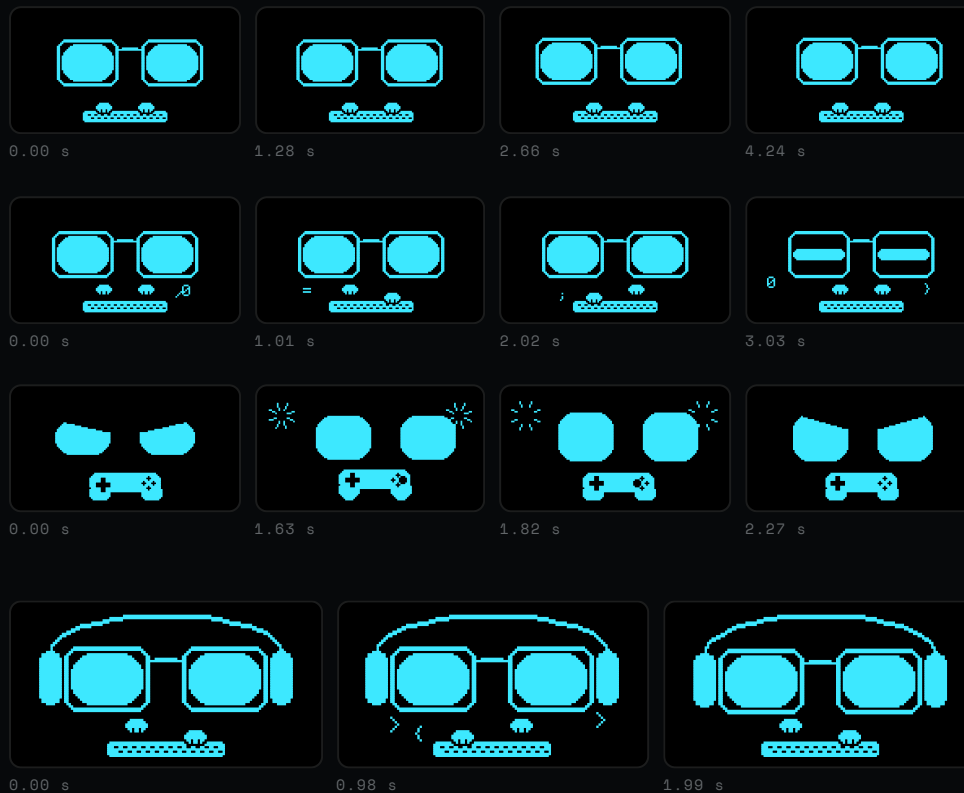
Aminal decides from the app and the window title, never from the sound itself.

Spotify is music. A browser tab that says Netflix is a film. A video player showing an .mp3 is music. Anything it can't name for certain, such as a video call or a YouTube video, gets no reaction at all. Games are noticed a different way (chapter 06).

- A new kind of sound has to play for 2 seconds before the pet reacts, so a notification chime is not a song.
- Silence has to last 6 seconds before it stops, so the gap between two tracks doesn't end the dance.
- When several apps make sound, the loudest one it can name wins, so a chime during a film doesn't take the popcorn away.
- Without beats from the PC, it dances to a steady 115 beats a minute of its own.

CODING AND GAMING

The pet also follows what's in front on the PC. With an editor or a terminal in front it puts on its glasses; with a game in front it picks up a controller.



Reading code

Glasses on, reading along a line, a quick hop back, and on down the page.

Typing

While the keys or the mouse are moving, two paws tap on a little keyboard and bits of code fly off its ends.

Gaming

Narrowed eyes dart after the action while the buttons get mashed. A loud moment in the game makes sparks burst out and the eyes pop wide (second and third frames).

With music on too

The headphones go on over whatever it's doing, and it nods on the beat.

HOW IT DECIDES

- **Code:** VS Code, Cursor, the JetBrains IDEs, Android Studio, Sublime Text, Notepad++, the Arduino IDE, any terminal, and GitHub, Colab or Jupyter in a browser.
- **Games:** anything run from a Steam, Epic, Riot, Xbox, EA, Ubisoft, GOG or Rockstar library (the launchers themselves don't count), a few games by name, and anything in exclusive full screen.
- It goes by the program in front, its folder and its window title. It never looks at what's in a window or what you type.
- Something has to stay in front for 3 seconds before it counts, so alt-tabbing past an editor isn't coding, and be gone for 8 before it ends, so a look at the docs isn't the end.
- An editor nobody has touched for 5 minutes isn't coding. A game never times out that way, because Windows doesn't count a gamepad as input.

SELFIES

Ask Amino for a selfie and the pet turns into a camera. While the webcam opens and meters for your face, the lens focuses and searches for you. As Amino says "Hold still, three, two, one", the numbers count down inside the lens. The moment the photo is taken, the screen flashes white and the shutter blinks shut and open.



Getting ready

"smile please!" while the camera opens, then 3, 2, 1 in time with the voice.

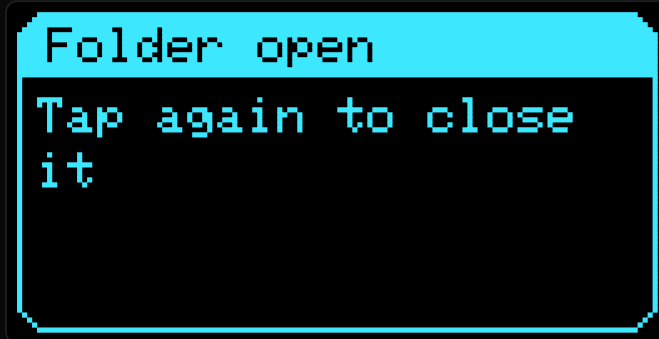
The shot

"cheese!", the flash, and the shutter closing. Then "click!", and "developing" until Amino has the picture.

The flash is the real shutter, not a timer. The photo burst starts the instant "three, two, one" finishes, and the bridge flashes the pet at exactly that moment, when Amino stops speaking. A guess would miss: the voice takes anywhere from 1.9 to 2.4 seconds to say it.

THE CARD

An RFID reader hides inside the right side of the pet's head (chapter 13). Tap the white card from its kit on the right side, near the front, and a hidden folder opens on the PC. Tap again and it closes.



Folder open

The pet looks up in surprise, says what it did on a card, and the folder opens.

- The first card ever tapped becomes the key. Any other card gets a sad face and "Not my card".
- The folder is called Vault and lives in your user folder. It's made on the first tap and marked hidden and system, so it stays out of sight even with hidden files shown. Hidden isn't locked: anyone who types the path can open it.
- A tap is the card arriving after being away for 1.5 seconds, so a card left lying on the reader counts once.
- It works with Aminimal closed: the bridge does it.

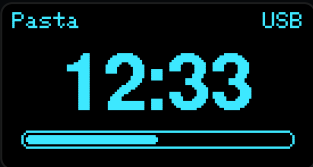
SCREENS AND CARDS



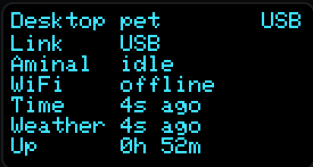
Clock
The time in big digits, the date above it, and a bar that fills across each minute.



Weather
An animated icon, the temperature, today's high and low, and the place. After three hours without news it adds "(old)".



Timer
Counts down by itself between updates, over a bar that drains. Paused, it blinks like an oven.



Status
The link, what Aminal is doing, WiFi, how fresh the time and weather are, and how long it has been up.

WEATHER ICONS

Every icon moves: the sun's rays turn, rain and snow fall, fog drifts and lightning flickers. From 7 pm to 6 am a clear sky shows a moon with twinkling stars. The numbers are the WMO weather codes Open-Meteo reports.



Clear
Code 0



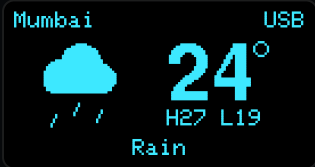
Partly cloudy
Codes 1 and 2



Overcast
Code 3



Fog
Codes 45 and 48



Rain and showers
Codes 61 to 67 and 80 to 82; drizzle (51 to 57) falls lighter



Snow
Codes 71 to 77, 85 and 86



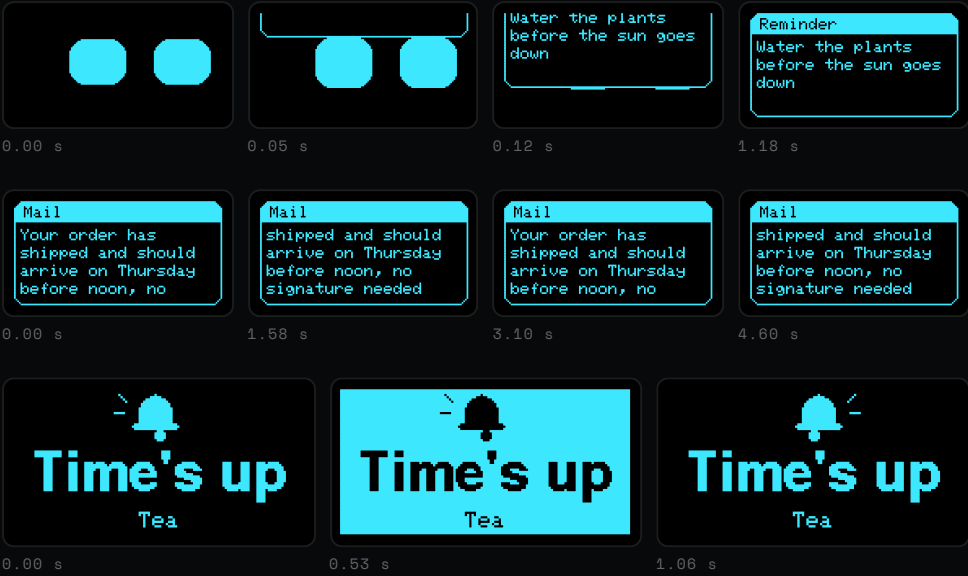
Thunderstorm
Codes 95 to 99



Clear, at night
Code 0 from 7 pm to 6 am

CARDS

Reminders, mail and what's next on the calendar slide down over whatever is on screen and stay for a few seconds. A tap puts one away early.



A card

Slides down from the top in 150 milliseconds.

A long card

Four lines fit. Longer text scrolls up a line every 1.5 seconds.

Time's up

When a timer runs out, the whole screen rings: a swinging bell, the timer's name, and a flash every half second until you tap it. The flash is the display inverting itself, which a frame-buffer capture can't see, so it is added back here.

THINGS TO TRY

Some of what the pet does only shows up when you go looking for it.

WITH YOUR FINGER

Pet it three times, fast.

Taps less than 1.5 seconds apart add up, and the third turns its eyes into hearts.

Leave it alone.

After 30 seconds it droops off to sleep. One tap wakes it.

Check on it late at night.

After 11 pm it falls asleep after only 10 seconds and shows the time while it sleeps, so it works as a night clock.

Double tap.

It walks through the clock, the weather, the timer and the status screen, then comes back to the face by itself.

Hold it on the timer.

The timer pauses and blinks. Hold again to carry on.

Tap your card on its right cheek.

A hidden folder opens. Tap again and it closes.

WITH AMINAL

Talk while the clock is up.

It flips back to the face to listen.

Speak up.

The eyes grow with your voice.

Ask for a selfie.

It counts down with the voice and flashes at the exact moment of the photo.

Close Aminor.

The pet keeps following your music, code and games. Only the listening, thinking and speaking stop.

Run it from a charger.

The bridge finds it over WiFi within about 30 seconds. Plug the cable back in and it moves back to USB by itself.

WITH THE PC

Play a long playlist.

The mood changes every 25 to 45 seconds: headphones, dancing, or the visualizer.

Watch the headphones.

Every nine seconds or so it peeks out.

Skip a track.

A gap shorter than 6 seconds doesn't stop the dance.

Get a message during a film.

The chime doesn't take the popcorn away.

Open VS Code and type.

Glasses on, paws on the keys, code flying off the keyboard.

Code with music on.

The headphones go on over the glasses.

Start a game.

It picks up a controller. Loud moments make sparks.

Join a video call.

Nothing happens, on purpose.

SMALL DETAILS

The bar under the clock

is the seconds, filling across the minute.

The sun's rays

turn, slowly.

Old weather

says so: after three hours without an update the screen adds "(old)".

A ringing timer

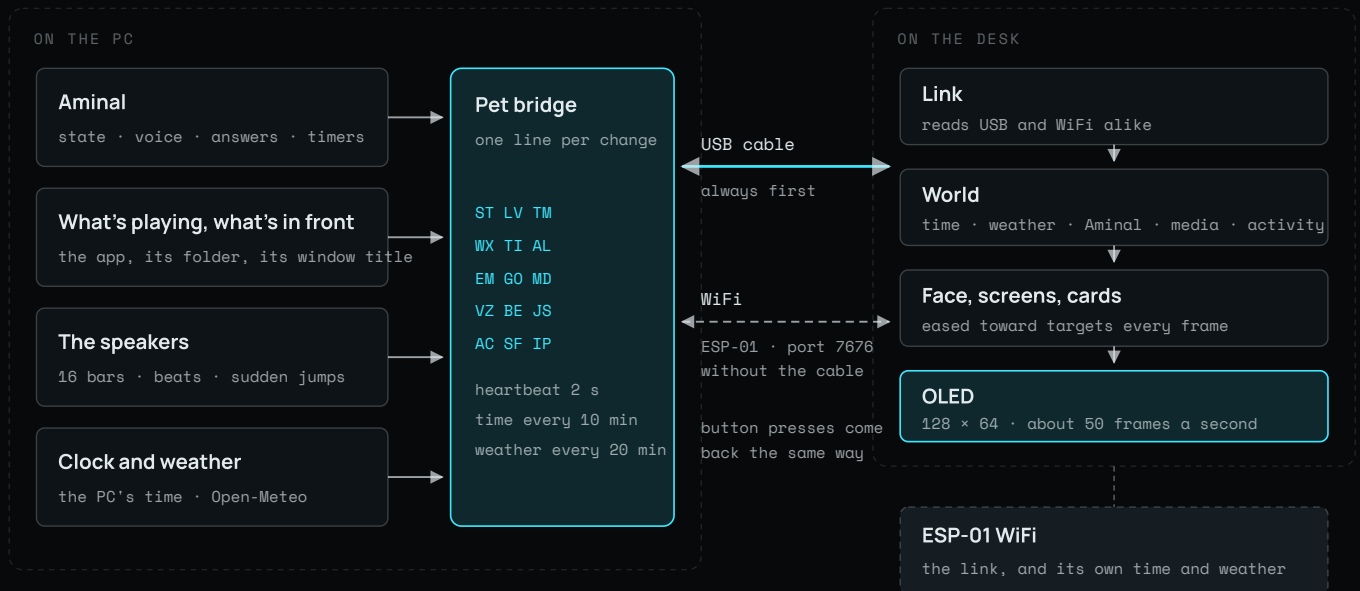
outranks everything: a reminder can't cover it.

Before anyone tells it the time,

the clock says "waiting for time" instead of guessing.

HOW IT FITS TOGETHER

Aminal never talks to the pet itself. A small bridge program reads what Aminal is doing 25 times a second, and what's playing and what's in front of you once a second, and turns every change into one short line of text. The pet reads those lines, keeps what they say in one shared state it calls the World, and draws that state 50 times a second.

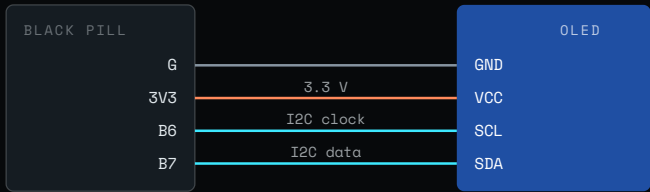


FINDING THE PET

- The bridge starts when you log in and runs whether Aminal is open or not. Only one copy ever runs; Aminal starts it if it isn't running, and never stops it.
- It only talks to a port that answers HI with PET, so it never sends lines to the wrong device.
- USB always wins. Without the cable it tries WiFi every 15 seconds, at the address the pet gave last time it was plugged in. If that doesn't answer, it searches the home network for the pet's port 7676, at most every 2 minutes.
- It sends a heartbeat every 2 seconds. After 8 seconds of silence the pet decides it is on its own and drops any music, code or game face. Over WiFi the bridge drops a pet that hasn't answered for 20 seconds, since a pet that lost its power cord leaves the connection looking open.
- The card is handled here too: the pet reports a tap, and the bridge opens or closes the folder.
- Nothing is sent twice: each kind of line goes out only when its value changes, apart from the time, which is refreshed every 10 minutes.

WIRING

PART	WHAT IT'S FOR
WeAct Black Pill	The brain: an STM32F401CC or F411CE, USB-C, the KEY and NRST buttons, and a blue LED on PC13.
0.96" SSD1306 OLED	The face: 128 × 64 pixels over I2C, at address 0x3C.
ESP-01	WiFi: the link to the PC without a cable, and its own time and weather.
RC522	Optional. The RFID card reader.
TTP223	Optional. A touch pad under the top of the head.



The OLED

Powered from the board's own 3.3 V. It draws little, so that is fine.



The RC522

Eight pins along one edge; IRQ stays empty. MISO and MOSI are the easiest pair to swap. The pet drives RST itself: left floating, the chip can sleep and stop answering.



The ESP-01

Eight pins in two rows of four. With the chips facing you and the antenna up, one row holds GND, GPIO2, GPIO0 and RX, the other TX, EN, RST and 3V3. GND and 3V3 sit in opposite corners.

POWER, AND SHORT WIRES

Three 3V3 and three G pins (two on the long headers, one of each on the small 4-pin header): one pair each for the OLED, the ESP-01 and the RC522. EN goes to A8, which the pet holds high, and the touch pad runs off B14 and B15 with its output on B0 (not A0, which the bootloader reads). Both 5 V pins stay empty: the ESP-01 and the RC522 are 3.3 V parts. Keep the ESP-01's wires short: its radio pulls 300 mA bursts, and long breadboard jumpers made it brown out.

THE HEAD

A 3D-printable head, $66 \times 89.5 \times 58$ mm: a soft cube with the screen behind a black visor, the Black Pill up on a stand inside, the touch pad under the top and the RFID reader inside the right side. Three parts, printed without supports. The STL files, print settings and how to put it together are in the **enclosure** folder of the repository.



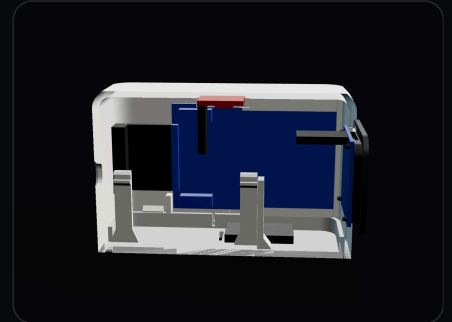
The head

The screen behind the visor.



The right side

Smooth on purpose: nothing shows where the card goes.



Inside the right wall

The RC522, antenna end at the front, its plugs toward the back.

- The reader hangs without screws: its top edge in a slot under the roof, its bottom edge in a groove whose inner side slopes at 45° . Tilt it, push the top into the slot, swing the bottom in and let it drop. A card reads through about 6 mm of plastic.
- The head was 70 mm deep until the reader went in. The board, its plugs and room for the wires to turn have to fit between the round corners, so the depth is worked out from them.
- The Black Pill sits 23.5 mm up, held only by its edges on four springy posts, so every jumper plug hangs free underneath.
- The touch pad senses your finger through 1 mm of the top.

THE FIRMWARE

C++ on the Arduino framework (STM32duino), built with PlatformIO. Adafruit's SSD1306 and GFX libraries draw the screen, and FreeSans Bold sets the big digits.

ONE PASS OF THE LOOP

- 01 Read every line waiting on USB and WiFi, up to 512 bytes a port per pass, so a flood of lines can't starve the animation.
- 02 Move the WiFi module's state machine on by one step: what the bridge said, what the pet answers, the weather request. It never waits for the module.
- 03 If 20 ms have passed since the last frame: read the button and the touch pad, ask the card reader whether a card is there (every 60 ms), check the link and the timer, update the face, draw the screen and any card, and send the 1,024-byte frame to the OLED over I2C.

That comes to about 50 frames a second, as fast as the I2C bus carries them.

EASING

No expression is drawn directly. Every frame, the eyes' position, width and height move part of the way toward a target:

```
current += (target - current) * 0.25 // position, width, and most heights
height += (target - height) * 0.6 // blinking, or closing by more than 10 px
height += (target - height) * 0.45 // while listening, to keep up with the voice
```

So glances glide, blinks snap shut, and falling asleep is a slow droop. Every face in this handbook is just a different set of targets.

WHO GETS THE FACE

The first of these that applies decides what the eyes do. Everything below Aminimal steps aside while it is listening, thinking or speaking, and music only adds its headphones when code or a game is in front.

- | | | |
|---|----------------------------------|----------------------|
| 1 A selfie: the whole screen becomes the camera | 2 A face Aminimal asked for | 3 Happy, after a tap |
| 4 Listening | 5 Thinking | 6 Speaking |
| 7 Code or a game in front | 8 Music | 9 A film |
| 10 Asleep | 11 Idle: glances, blinks, habits | |

TIMINGS

WHAT	WHEN	WHAT	WHEN
Glance	every 1.5–4 s	Music mood	25–45 s
Blink	every 2.5–6 s, 130 ms	Its own tempo	115 bpm, 520 ms
Habit	every 15–40 s	Popcorn	every 3.5–7 s

WHAT	WHEN	WHAT	WHEN
Happy after a tap	2.5 s	Back to the face	after 20 s
Hearts	3 taps under 1.5 s apart	Link lost	8 s of silence
Falls asleep	30 s; 10 s, 11 pm–7 am	Hold / double tap	0.7 s / 0.35 s
Code or a game	3 s in, 8 s out	Next card tap	after 1.5 s away

Every timer is compared with `reached(now, t)` rather than `now > t`, so the pet keeps blinking when the millisecond counter wraps round after 49.7 days.

THE PROTOCOL

One short line of text per message, the same on USB and WiFi, and simple enough to type into a serial monitor. A line longer than 159 characters is dropped whole rather than read in pieces.

LINE	MEANING
HI	Who are you? The pet answers PET desktop-pet 10.
PG	Heartbeat, answered P0.
ST idle listening thinking speaking off	What Aminimal is doing
LV 0-100	How loud the voice is, for the eyes and the mouth
TM <unix> <offset s>	The time, and the UTC offset
WX <temp> <code> <hi> <lo> <place>	The weather, with its WMO code
TI <left> <total> <paused> <label>	The timer, or TI - for none
AL <secs> <title> <text>	A card, or AL - to take it down
EM <face> [ms]	Pull a face: happy, love, surprised, sad, angry, wink, sleepy
G0 <screen> [secs]	Show face, clock, weather, timer or status
MD music [style] watch -	What's playing. A music style (headphones, dance, bars) can be asked for by name.
VZ <16 hex digits>	Spectrum bars, 0 to f each, low notes first
BE	A beat
JS	A sudden loud moment in a film or a game
AC code [typing] game -	What's in front on the PC
SF ready count <ms> shot -	A selfie: the camera opening, the spoken countdown, the shutter, done
SN	Answered SN <hex>: exactly what is on the OLED
IP	Answered IP <address> or IP -: where it is on WiFi
RF, RF test, RF poll	The card reader: its chip version, a wiring test, one try at reading a card
L0 <lat> <lon> <place>	Where the weather is for, kept in flash for the WiFi
WF <ssid><tab><password>	USB only: join a WiFi network
ES talk flash [baud], ES reset	USB only: the cable straight through to the ESP-01, for debugging

The pet sends events back up whichever link spoke last: EV pet, EV next <screen>, EV hold, EV hold timer, EV dismiss, EV wifi ok <ip>, EV wifi fail and EV card <uid> tap.

A FEW SECONDS OF CONVERSATION

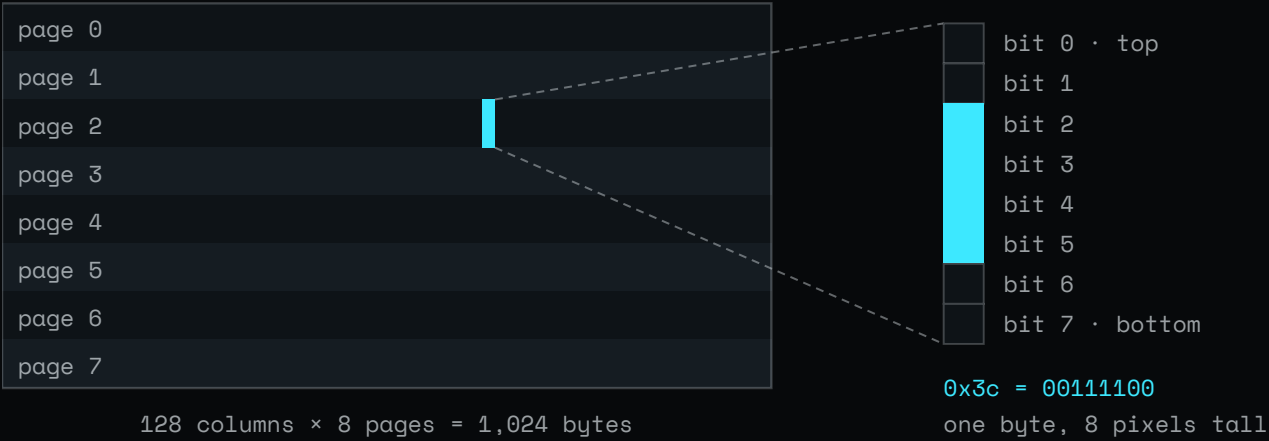
```
→ HI
← PET desktop-pet 10
→ TM 1790658725 19800           // 10:42 on 29 Sep, India's +5:30
→ WX 29 2 32 24 Mumbai
→ ST listening
→ LV 64
→ ST speaking
→ MD music dance
→ BE
→ AC code typing               // VS Code in front, keys going
← EV pet                       // someone tapped it
```

SNAPSHOTS: HOW THESE PICTURES WERE MADE

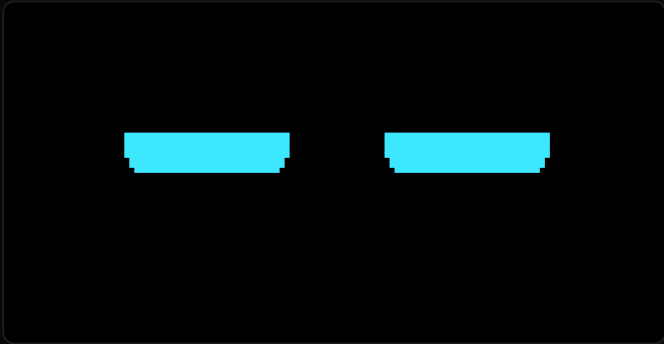
SN answers with the whole frame buffer as 2,048 hex digits. The buffer is 8 pages of 128 bytes, and each byte is a column of 8 pixels with the top pixel in the lowest bit. Here is the start of a real reply, taken while the pet was in love, and a piece of page 3, where the hearts are:

```
SN 0000000000000000000000000000000000000000000000000000000000000000... // page 0: the top 8 rows, dark
... page 3, from column 24:
ffffffffffffffffffffffffffffffffffffffffffffffffffffffffffff7f1f0000000000000000
... 2,048 digits in all
```

Over USB one snapshot takes about 36 ms, so the pet can be filmed at 28 frames a second while it keeps drawing its own 50. Over WiFi one takes about 0.4 seconds. Each strip in this handbook is a run of those frames, decoded and drawn back pixel for pixel.



Decoded
The reply above, drawn back: all 8 pages, 128 × 64 pixels.



Page 3 alone
Only bytes 384 to 511, the 8 rows through the middle of the hearts.

HOW IT HEARS MUSIC

All of this runs on the PC, inside the bridge. The pet only ever receives the results as MD, VZ, BE and JS lines.

STEP	WHAT HAPPENS
1 · What's playing	Once a second, Windows' per-app volume meters list the apps making sound (a peak above 2%) . Each is named by its app, then its window title, then the file it plays. The loudest one that gets a name wins.
2 · Listening in	Only while music or a film is on, or a game is in front, the bridge records what the speakers play through WASAPI loopback (PyAudioWPatch).
3 · Bars	Each block of 1,024 samples, about 23 ms, goes through an FFT into 16 bands spaced by pitch from 50 Hz to 12 kHz. The top end is lifted 4.5 dB per octave so the treble moves as much as the bass, and the bars show the loudest 40 dB under an automatic gain. Sent as VZ 20 times a second.
4 · Beats	Energy between 40 and 150 Hz rising to 1.45 times its average over the last second, at least 0.28 s after the last beat. Sent as BE.
5 · Jumps	A block 4 times louder than the median of the last 5 seconds, at most once every 8 seconds. Sent as JS, for films and games.

THE SOUND STAYS PUT

Each block becomes 16 numbers and two yes-or-no flags and is thrown away. Nothing is recorded, saved or sent anywhere, and the listener only runs while the pet has music, a film or a game to show.

NAMED FOR CERTAIN, OR NOT AT ALL

CLUE	MUSIC	FILM
The app	Spotify, Apple Music, Tidal, Deezer, foobar2000, MusicBee...	Netflix, Prime Video, Disney+, Hotstar, JioCinema apps
A browser's title	Spotify, YouTube Music, SoundCloud, JioSaavn, Gaana, Wynk	Netflix, Prime Video, Hotstar, JioCinema, Disney+, SonyLIV, ZEE5, Apple TV
A video player's file	.mp3, .flac, .m4a, .wav, .ogg, .opus, .aac, .wma	anything else it plays

Browsers are recognised even when they aren't on the list, by the audio helper process every Chromium and Firefox browser uses. That is how Comet was picked up.

WIFI

The ESP-01 keeps the factory AT firmware it shipped with, and the pet drives it with plain text commands. Once it has joined a network it makes one plain HTTP request to Open-Meteo every 20 minutes. That single reply carries the time (in its Date header), the UTC offset and today's weather, so a pet on a charger keeps its clock and forecast with the PC switched off.

It also listens on port 7676, and whoever connects there, the bridge on the PC when the pet has only a power cord, speaks the same protocol as USB. The module's multi-link mode makes that possible: the weather request goes out on link 4 while the bridge comes in on another. While the bridge is connected it brings the time and weather itself, so the pet skips its own request.

```
start → reset → ATE0 → AT+CWMODE_DEF? → AT+CIPMUX=1 → AT+CIPSERVER=1,7676 → AT+CIFSR → idle
idle  → AT+CIPSTART=4,"TCP","api.open-meteo.com",80 → AT+CIPSEND=4,... → GET → +IPD,4,... → 4,CLOSED
      // every 20 min while no bridge is connected; again after 5 min if it failed
idle  ⇄ +IPD,0,... in from the bridge, AT+CIPSEND=0,... back to it
```

JOINING

Run the bridge's WiFi setup with the pet on USB. It offers the network the PC is on, asks for the password without showing it, and sends both down the USB cable only: the pet refuses them over WiFi. The module remembers the network and rejoins by itself. It only sees 2.4 GHz networks.

WHAT IT KEEPS

Where the weather is for (sent by Amino as Lo) and the UTC offset are saved in the last sector of the Black Pill's flash, so they survive a power cut. The WiFi password isn't: only the module holds it.

A CLEAN START

The pet resets the module at every power-up, and again whenever it stops answering, so a bad power-up can't leave the WiFi stuck. Anyone on the same network can reach port 7676 too, so it's meant for a home network; joining a network and the passthrough to the module stay USB only.

BUILDING AND UPLOADING

WeAct boards ship with the WeAct HID bootloader in their first 16 KB. Windows sees it as a plain HID device (0483:572A), so it needs no driver, and it avoids the chip's own USB bootloader, which often fails with Device Descriptor Request Failed.

- 01 **Install PlatformIO.** In VS Code, open Extensions, search for PlatformIO IDE and install it.
- 02 **Get the uploader.** Download `WeAct_HID_Flash-CLI.exe` from WeAct's repository and save it as `tools/WeAct_HID_Flash-CLI.exe`. It is WeAct's own closed tool, so it isn't in the repository.
- 03 **Open the project.** File, Open Folder, `Desktop-pet`. PlatformIO fetches the STM32 toolchain and the display libraries itself.
- 04 **Enter the bootloader.** Hold KEY, tap NRST, wait for the blue LED to blink, then let go of KEY. A brand new board is already in it.
- 05 **Upload.** Click the arrow in the blue status bar.
- 06 **Wake it.** The eyes appear. If they don't, tap NRST.

The default build targets the F401CC and also runs on an F411CE. For the F411's full 100 MHz, check the square chip and switch `default_envs` to `blackpill_f411ce`.

WHERE EVERYTHING LIVES IN FLASH



That is the F401CC's 256 KB, drawn to scale. The app is linked to start right after the bootloader, at 0x08004000, and has about 8 KB of room before it would meet the saved settings. The build refuses to grow past them: PlatformIO is told where flash ends for the app.

WHEN SOMETHING'S OFF

WHAT YOU SEE	LIKELY CAUSE
Blue LED blinking fast, screen blank	The OLED isn't answering on I2C. Check its four wires, then try address 0x3D in <code>src/main.cpp</code> .
Upload can't find the board	It isn't in the bootloader. Hold KEY, tap NRST, let go when the LED blinks.
No HID device at all, even holding KEY	The bootloader was erased by a DFU or ST-Link upload to 0x08000000. Reflash it from WeAct's repository, or build without the flash offset.
Garbage on the screen	The module may be an SH1106, common on 1.3 inch boards, which needs a different library.
The pet never follows the PC	The status screen says Link none: the bridge isn't running (it starts at login), or something else has the port.
ESP-01 answers AT, then garbles and restarts	A brownout, usually from long jumper wires. Use short direct ones (chapter 12).
The card does nothing	Send <code>RF test</code> over USB. A value written to the reader should read back the same, <code>rw 5a/5a a5/a5</code> . If every read is the same number, a wire to the RC522 is wrong or loose; check SDA, SCK and RST first.

A MAP OF THE CODE

FILE	WHAT'S IN IT
<code>src/pet.h</code>	The shared World: time, weather, timer, Aminal's state, media, what's in front, a selfie, the current card
<code>src/main.cpp</code>	Setup, the frame loop, the button, switching screens
<code>src/face.cpp</code>	Eyes, moods, habits, and the listening, thinking, speaking, music and film faces
<code>src/media.cpp</code>	Headphones, music notes, the visualizer and the popcorn
<code>src/activity.cpp</code>	Glasses, the keyboard and paws, the controller and the sparks
<code>src/camera.cpp</code>	The camera face for a selfie: countdown, flash and shutter
<code>src/rfid.cpp</code>	The card reader, and turning a card's arrival into a tap
<code>src/screens.cpp</code>	Clock, weather icons, timer, status and the cards
<code>src/link.cpp</code>	The protocol, read from USB and WiFi alike, and the ESP-01 passthrough
<code>src/wifi.cpp</code>	The ESP-01 driven through its AT firmware: joining, the weather request, and the bridge's link on port 7676

DOCUMENT HISTORY

REVISION	DATE	WHAT CHANGED
Rev. 2	2 Oct 2026	For firmware v10: coding and gaming faces, the selfie camera, the RFID card, WiFi as the link to the PC, a bridge that runs without Aminal, Bluetooth taken out, new wiring for the ESP-01, the RC522 and the touch pad, and a chapter on the printed head.
Rev. 1	29 Sep 2026	First edition, for firmware v6: the user guide, the technical notes, and real captures of every face and screen.